

ARYAN RAMESH SURANI

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EDUCATION

Sardar Vallabhbhai National Institute of Technology
Bachelor of Technology in Computer Science (CGPA: 6.77)

Surat, Gujarat
Aug. 2024 - Present

Birla Public School, Doha
Higher Secondary Education (90.89%)

Doha, Qatar
Jun. 2022 - Apr. 2024

Birla Public School, Doha
Secondary Education (91.42%)

Doha, Qatar
Apr. 2020 - Mar. 2022

PROJECTS

ClusterFolio - Systematic Portfolio Construction & Backtesting Engine | *Python, pandas, NetworkX, scikit-learn, Streamlit*

- Built a 12-stage quant pipeline spanning 12 global equity indices (NIFTY50, S&P500, NASDAQ100, DAX, etc.), clustering stocks via correlation graphs and Louvain community detection before forming basket portfolios.
- Implemented two-layer capital allocation: 6 basket-level strategies (Equal Weight, Markowitz, Risk Parity, Black-Litterman, etc.) combined with 5 independent within-basket strategies grounded in published finance literature (George & Hwang 2004; Lee & Swaminathan 1998).
- Built an ML ensemble scorer (Decision Tree + Random Forest + Gradient Boosting) validated via an 80/20 chronological walk-forward split, alongside a deterministic hardcoded-scoring alternative.
- Designed a whole-share capital-sizing module computing the minimum capital needed to hold every selected stock near its target weight, addressing NSE/BSE's no-fractional-share constraint.
- Shipped a walk-forward backtesting engine (monthly rebalancing, transaction costs, CAGR/Sharpe/Drawdown/Turnover stats) with a Streamlit dashboard for equity-curve comparison, backed by a 30-test pytest suite and a Ubuntu/Windows CI pipeline; example result: KOSPI 200 with ML scoring + Black-Litterman achieved 33.4% CAGR / 2.19 Sharpe vs. a 27.0% CAGR benchmark.

GitHub: github.com/ars231106/Basket-Intelligence-System

EnduranceHub - Full-Stack Endurance Training Analytics Platform | *FastAPI, PostgreSQL, JWT, Docker*

- Built a full-stack analytics platform (~5,000 lines, 22 REST endpoints) supporting 28 sports, computing training load, injury-risk ratios (ACWR), monotony/strain, streaks, and personal records.
- Implemented secure authentication (bcrypt hashing, JWT sessions, email OTP verification) with enumeration-safe error handling and a soft-delete account lifecycle with a scheduled purge job.
- Delegated password storage to the OS keychain via the browser's Credential Management API for silent re-login, avoiding any in-app handling of raw credentials.
- Designed independent per-sport weekly goal tracking with per-sport progress bars and a combined completion ring averaging across distance- and duration-based sports.
- Built a framework-free JavaScript frontend (landing page + 4-section dashboard) with Chart.js visualizations, a 13-week load heatmap, and a pre-paint System/Light/Dark theme; deployed via Docker to Render with a 5-job CI pipeline and a ~40-test suite.

GitHub: github.com/ars231106/Endurance-Hub

EdgeAudit - ML Market Signal Research & Monte Carlo Risk Engine | *Python, scikit-learn, XGBoost, LightGBM*

- Built a modular Python pipeline (data ingestion, feature engineering, model selection, prediction, Monte Carlo, backtesting, reporting) for per-ticker market analysis.
- Engineered technical features (SMA, RSI, ATR, momentum, volume-change) and cross-validated five classifier families (Logistic Regression through LightGBM) using a purged/embargoed time-series split to prevent lookahead leakage.
- Ran a systematic, multi-week test across labeling schemes (fixed-horizon, triple-barrier, volatility-direction) and concluded price-direction prediction carries no reliable edge (AUC 0.48-0.58) while volatility-direction prediction shows a real, replicated edge (AUC 0.65-0.75) across four tickers.
- Built a Monte Carlo simulation engine producing expected/median/90%-range price forecasts with static and animated visualizations.
- Built a rules-based paper-trading backtester (70/30 split, 0.55 confidence threshold) compounding a simulated \$100,000 portfolio against buy-and-hold, then compiled results into an automated per-ticker market intelligence report.

GitHub: github.com/ars231106/Stock-Trend-Predictor---Monte-Carlo-Risk-Engine

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, JavaScript, HTML/CSS

Frontend: Vanilla JS, Chart.js, Streamlit

Backend: FastAPI, SQLAlchemy, PostgreSQL, JWT/OAuth2

ML & Quant: scikit-learn, XGBoost, LightGBM, NetworkX, pandas, NumPy

Developer Tools: Git, GitHub Actions (CI/CD), Docker, pytest, Ruff

ACHIEVEMENTS

JEE Main 2024: Achieved 86 percentile with a CRL rank of 2,11,310.

Competitive Programming: Solved 250+ problems across LeetCode and NeetCode, reinforcing core data structures and algorithms.